

BridgeEDI — Integration Architecture (Client Draft)

Platform: BridgeEDI
Purpose: Suggested architecture for ERP ↔ Government / Third-party integration

1. One-line pitch

BridgeEDI sits between the customer ERP and external systems (government portals, partner APIs). It sends and receives messages, calls APIs, transforms data, monitors everything, and uses AI to explain failures.

2. Simple architecture diagram

```
flowchart LR
    ERP["Customer ERP"]
    BRIDGE["BridgeEDI Platform"]
    GOV["Government Portal / APIs"]
    TP["Third-party APIs"]
    AI["AI Services"]
    MON["Monitoring Portal"]
    ERP <-->|Send / Receive messages| BRIDGE
    BRIDGE <-->|Submit / Callbacks| GOV
    BRIDGE <-->|API calls| TP
    BRIDGE --> AI
    BRIDGE --> MON
    AI --> MON
```

What is inside BridgeEDI (simple)

```
flowchart TB
    ERP["Customer ERP"]
    GW["API Gateway<br/>Secure entry"]
    CORE["Process<br/>Validate · Transform · Rules"]
    CONN["Connectors<br/>Government · ERP · Partners"]
    STORE["Logs · Database · Files"]
    MON["Monitor · Alerts · AI"]
    EXT["Government / Third-party"]
    ERP --> GW
    GW --> CORE
    CORE --> CONN
    CONN --> EXT
    EXT -->|Callbacks| GW
    CORE --> STORE
    STORE --> MON
    CONN --> MON
```

3. How a transaction works (quick)

```
Customer ERP ↓ BridgeEDI (receive + validate + transform) ↓ Government / Partner API ↓
BridgeEDI (response + update ERP) ↓ Monitoring + Audit + Alerts
```

If it fails: retry → log → notify user.

4. What BridgeEDI does

Capability	Meaning
Send / receive messages	Talk to ERP and partners

Call APIs	Government + third-party
Receive webhooks	Async status callbacks
Transform & validate	ERP format ↔ external format
Monitor	See success, failure, latency
Audit	Full trail of every request
AI assist	Explain failures, search in plain language

5. AI role (simple)

AI helps operations — it does **not** replace the integration.

- “Why did this fail?”
- “Show failed submissions today”
- Alerts when error rates go up
- Short ops reports

6. Why this approach

- One platform instead of many point-to-point links
- ERP does not need to know government API details
- Secure, logged, retryable
- Easy to add more customers / agencies later

BridgeEDI — Integration. Visibility. Trust.
Draft for discussion — not a final build plan.